

Website UI Redesign Report

IRCTC Website Redesign (UI/UX Case Study)

Submitted By: Mahima Singh

Internship Domain: UI/UX Design

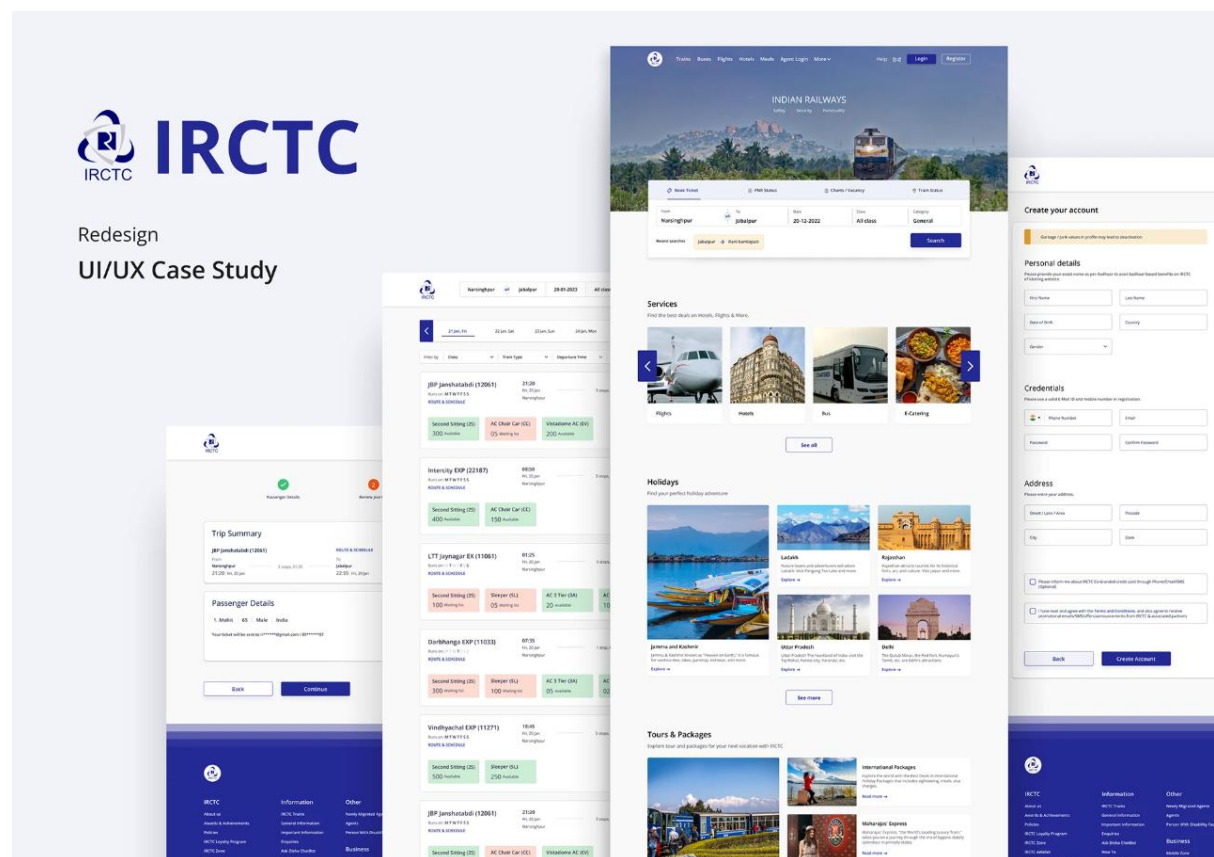
Tools Used: Figma, Adobe Photoshop, Google Forms

1. Introduction

The Indian Railway Catering and Tourism Corporation (IRCTC) is India's largest railway ticket booking platform. Millions of users visit the website every day to book train tickets, check PNR status, reserve hotels, and explore tourism services.

Despite offering numerous services, the existing website suffers from several usability issues including cluttered navigation, inconsistent layouts, poor visual hierarchy, and an outdated interface.

This project focuses on redesigning the IRCTC website with modern UI/UX principles to improve user experience while maintaining the website's functionality.



2. Project Overview

Project Name

IRCTC Website UI Redesign

Platform

Website (Desktop + Responsive)

Duration

2 Months

My Role

Sole UI/UX Designer

Tools

- Figma
- Adobe Photoshop
- Google Forms

Project Overview

Indian Railway Catering and Tourism Corporation (IRCTC) is an Indian Railway subsidiary established by the Ministry of Railways to manage Indian railway catering, online ticketing and tourism operations.

On 19 March 2014, nearly **5,80,000** tickets booked in a day.

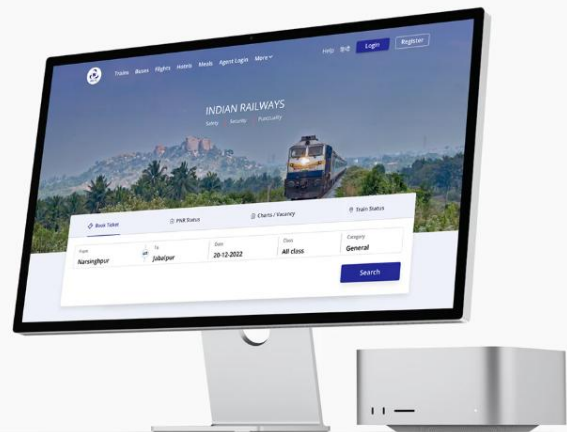
On 1 April 2015, **13,45,496** tickets booked in a day.

On 2 April 2015, nearly **11,00,000** tickets booked in a day.

On April 2015, nearly **13,40,000** tickets booked in a day

Project Goal

The goal of the project was to redesign the layout, hierarchy, flow, and navigation and make the design intuitive and easy to use according to the latest trends.



My Role

Sole UI/UX Designer

Timeline

2
Months

Tools



3. Problem Statement

During the evaluation of the existing IRCTC website, several usability issues were identified:

- Cluttered homepage with excessive information
- Difficult navigation structure
- Poor visual hierarchy
- Multiple competing call-to-action buttons
- Outdated design language
- Inconsistent spacing and typography
- Poor mobile responsiveness
- Ticket booking process required unnecessary effort

These issues reduce usability and increase the cognitive load for users.

Problem Statement

- Inconsistency, Bad layout, Lack of hierarchy, Random placement of CTAs.
- Advertisements were very close to the CTAs, causing users trouble finding and using the CTAs.
- Users were unable to select the seats and berths according to their preferences.
- Complex and lengthy registration process.
- Cluttered and outdated design without hierarchy creates confusion and prevents users from reaching useful actions and information.
- Captcha on every single page frustrated users and slowed down the website's performance.
- Bad navigation and lengthy process make the product complex, and the more complex it is, the more time users take to make decisions, which causes cognitive load.
- There was no option to get the running status of the train.

Solution Statement

- Provided proper layout, hierarchy, and fixed place for CTAs.
- Changed the placement of the advertisements so that they do not cause any trouble to the users.
- Added the drop-down menu from which users can select seats and berths according to their preferences.
- Provided simple and short registration process.
- Provided clean and minimal design with a proper hierarchy, which allows users to take actions or scan the information easily without any confusion.
- Only one captcha on the login page to differentiate between humans and computers for security purposes.
- Designed a simple user flow and short process to complete the task, which takes users less time, makes them take decisions fast, and reduces the cognitive load.
- Added the option to check the running status of the train.

4. Design Process

The project followed the Design Thinking Process.

Step 1 — Research

- Studied the existing IRCTC website
 - Observed user behavior
 - Identified pain points
 - Collected user feedback
-

Step 2 — Heuristic Evaluation

The website was evaluated using Nielsen's 10 Usability Heuristics.

Major findings:

- ✓ Inconsistent navigation
 - ✓ Lack of visual hierarchy
 - ✓ Information overload
 - ✓ Small clickable areas
 - ✓ Poor feedback after user actions
 - ✓ Excessive form fields
-

Step 3 — User Research

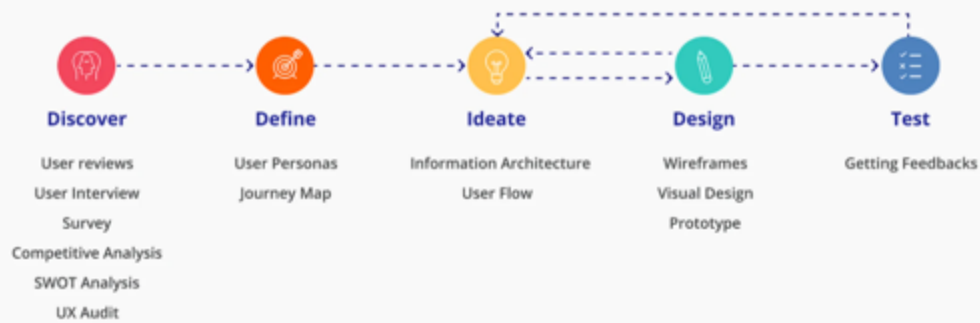
Methods Used:

- User Interviews
- Competitor Analysis
- Online Survey

Common user feedback:

- Booking feels confusing
- Homepage is overcrowded
- Navigation is difficult
- Ticket search should be faster

Design Process



Why I Redesign IRCTC?

I've been using IRCTC to book train tickets for many years, and during that time, I've encountered numerous issues with this website, such as

- Unable to log in due to the perplexing captcha design that is stuck with the advertisement
- It was necessary to log in even to search for the trains, and I had to enter my username and password to search for trains.
- There was a captcha on every single page, which was time-consuming and irritating.
- The user interface was cluttered and confusing.
- Booking a ticket with the IRCTC was like wasting 10 to 15 minutes because of that bad design, so I decided to redesign it.

Above are the only few pain points that I describe; there are a lot more. I was aware of the bad design of the IRCTC website, but I wanted to know what other users were feeling or saying about it, which is why I first decided to read the user reviews, and for that, I visited the [MouthShut](#) website.

5. User Reviews

User Reviews

Horrible experience

★☆☆☆☆ Oct 10, 2022 05:52 PM • 287 Views [via Mobile](#)

Extremely non user friendly, too many irritating advertising, lack of easy access to useful information
They are not improving - hopeless

Worst website for user registration and booking

★☆☆☆☆ Oct 30, 2021 08:39 PM • 1322 Views [via Mobile](#)

It says the mobile number is already in use, if I use another mobile number the captcha is not working
I have been trying for almost a dozen times in the last two years!

THE WORST

★☆☆☆☆ Apr 10, 2022 10:32 PM • 9344 Views

Very bad experience on this website, can't even register without breaking my laptop out of frustration, I would rather travel 50km and do it manually than to book from this site. What a shitty site

Customer service is very bad

★☆☆☆☆ Oct 10, 2022 08:29 AM • 527 Views [via Mobile](#)

Recently called irctc customer service, there is lot of hold time and one lady responds she is not even listening to our queries, just saying for RAC enquiry call 139. If that so, we are contacting customer service to know detailed information, not just like transferring to other phone number. Atleast in future expecting irctc customer service behave in a good manner

IRCTC WEBSITE

★☆☆☆☆ Dec 10, 2020 06:22 AM • 1286 Views

IT IS VERY tough to use the site, when I clicked the menu, it didn't display the menu items. How can I cancel my ticket?

Worst website

★☆☆☆☆ Oct 29, 2022 09:46 PM • 430 Views [via Mobile](#)

Looks like this website is created only for railways. Just does not work. Cannot create the user id. Not at all user friendly. No point in creating better platforms and faster trains as nobody can book tickets online

New IRCTC portal designer challenge on booking

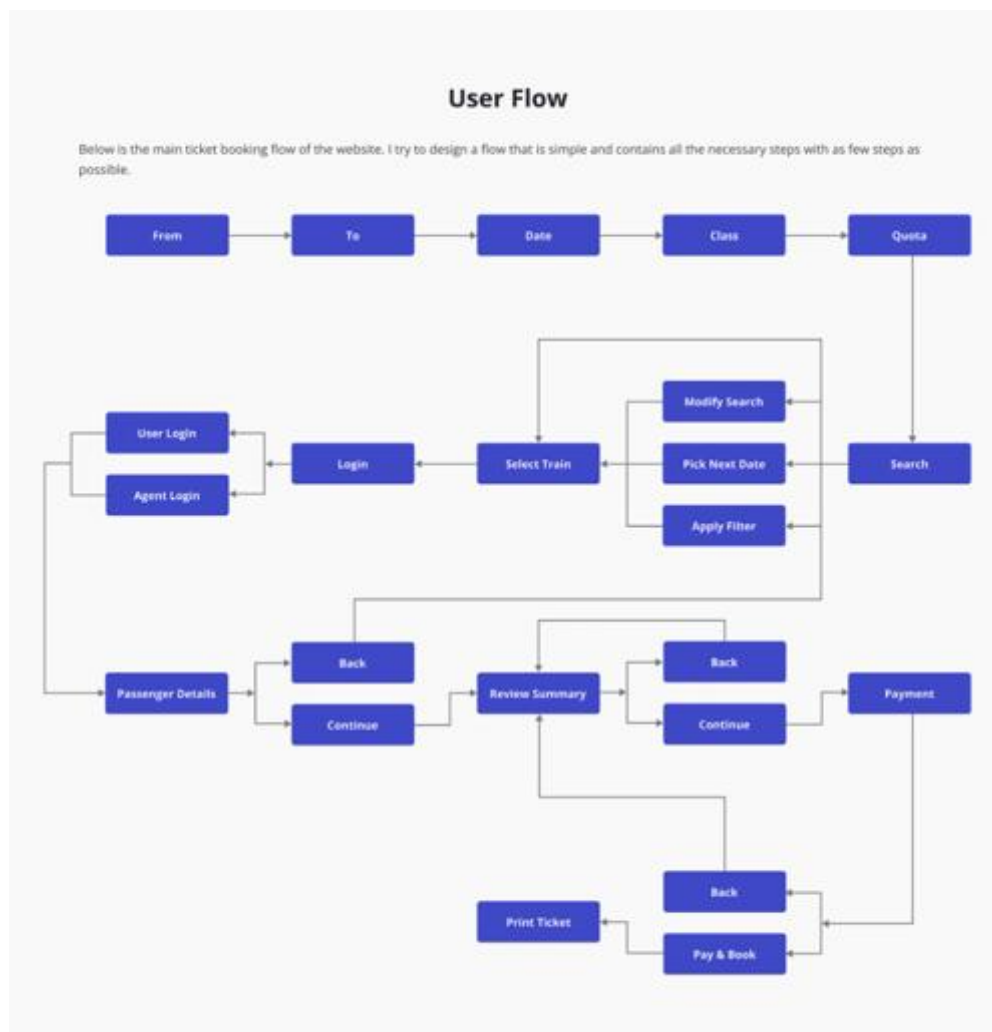
★☆☆☆☆ Aug 24, 2022 08:10 PM • 412 Views [via Mobile](#)

Super IRCTC website designer challenging customers to book a ticket on this portal if possible. I gave up I mean I rock this website. Today I spent so much time to book a ticket n gave up myself after getting tired with advertisements that to on payment page n could not complete payments at last. Kudos to ur thought men? thanks for making our health improve by visiting offline booking counters.

6. User Persona



7. User Flow



8. Comparison Summary

COMPARISON SUMMARY

CRITERIA	 IRCTC	 make my trip	 goibibo	 RAILYATRI	 ixigo
 Train Booking	✓	✓	✓	✓	✓
 Flight Booking	✗	✓	✓	✗	✓
 Hotel Booking	✗	✓	✓	✗	✓
 Bus Booking	✗	✓	✓	✗	✓
 Modern UI	★ ★ ☆ ☆ ☆	★ ★ ★ ★ ★	★ ★ ★ ★ ☆	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆
 Easy Navigation	★ ★ ☆ ☆ ☆	★ ★ ★ ★ ★	★ ★ ★ ★ ☆	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆
 Responsive Design	★ ★ ★ ☆ ☆	★ ★ ★ ★ ★	★ ★ ★ ★ ☆	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆
 Booking Experience	★ ★ ★ ☆ ☆	★ ★ ★ ★ ★	★ ★ ★ ★ ☆	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆
 Extra Travel Services	★ ★ ☆ ☆ ☆	★ ★ ★ ★ ★	★ ★ ★ ★ ☆	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆
 Trust & Reliability	★ ★ ★ ★ ★	★ ★ ★ ★ ☆	★ ★ ★ ★ ☆	★ ★ ★ ☆ ☆	★ ★ ★ ★ ☆

KEY INSIGHTS



IRCTC

Most trusted platform for railway ticket booking.
Needs improvement in visual design and navigation.



MakeMyTrip

Best overall user experience with a clean interface.
Offers complete travel booking services.



Goibibo

Easy-to-use interface with attractive travel deals.
Smooth booking experience.



RailYatri

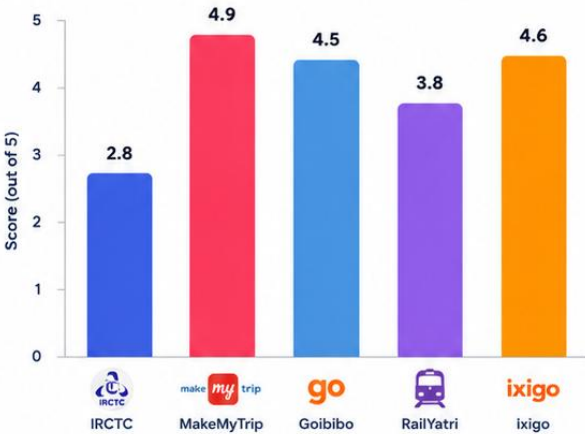
Excellent for railway-related information.
Limited travel services beyond trains.



ixigo

Modern design with a fast booking process.
Strong mobile experience and smart travel features.

OVERALL UX COMPARISON SCORE

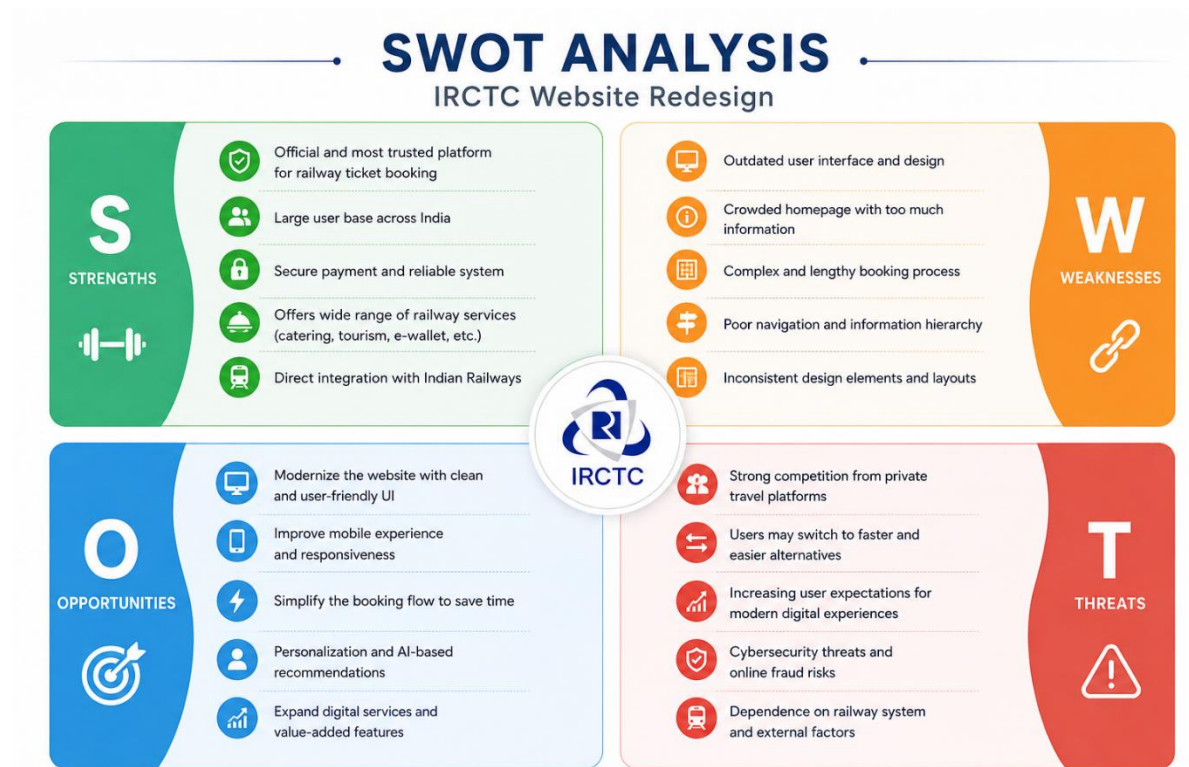


✓ = Available

✗ = Not Available

★ ★ ★ ★ ★ = Overall User Experience Rating

9. SWOT Analysis



10. Expected Outcome

This project helped me gain practical experience in:

- Improving website usability
- Designing intuitive user interfaces
- Creating responsive layouts
- Developing high-fidelity prototypes
- Applying UI/UX design principles to solve real-world usability problems

11. Conclusion

The IRCTC Website UI Redesign project successfully transformed a complex and outdated interface into a cleaner, more user-friendly, and visually modern experience. By applying user-centered design principles, heuristic evaluation, responsive design techniques, and interactive prototyping, the redesigned website improves navigation, simplifies the booking process, and enhances the overall user experience.

This project strengthened my understanding of the complete UI/UX design process—from research and analysis to high-fidelity design and prototyping—and provided valuable hands-on experience in solving real-world usability challenges.